

Database Technologies – Assignment 10

PL-SQL 1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$.)

```
KD3-98849-sakshi@localhost> DELIMITER $$
KD3-98849-sakshi@localhost> CREATE PROCEDURE reactangle()
-> BEGIN
-> DECLARE LENGTH INT DEFAULT 10;
-> DECLARE WIDTH INT DEFAULT 5;
-> DECLARE AREA INT;
-> DECLARE PERIMETER INT;
-> SET
-> AREA=LENGTH*WIDTH;
-> SET
-> PERIMETER=2*(LENGTH+WIDTH);
-> SELECT LENGTH,WIDTH,AREA,PERIMETER;
-> END;
-> $$
```

Query OK, 0 rows affected (0.167 sec)

```
KD3-98849-sakshi@localhost> DELIMITER ;
KD3-98849-sakshi@localhost> CALL reactangle();
```

LENGTH	WIDTH	AREA	PERIMETER
10	5	50	30

1 row in set (0.049 sec)

Query OK, 0 rows affected (0.090 sec)

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its

cube.

```
KD3-98849-sakshi@localhost> CREATE table temp(
-> value INT,
-> square INT,
-> cube_value INT
-> );
Query OK, 0 rows affected (0.415 sec)

KD3-98849-sakshi@localhost> DROP PROCEDURE IF EXISTS reactangle;
Query OK, 0 rows affected (0.408 sec)

KD3-98849-sakshi@localhost> DELIMITER $$
KD3-98849-sakshi@localhost> CREATE PROCEDURE reactangle()
-> BEGIN
-> DECLARE num INT DEFAULT 5;
-> INSERT INTO temp VALUES(num,num*num,num*num*num);
-> END;
-> $$
Query OK, 0 rows affected (0.066 sec)

KD3-98849-sakshi@localhost> DELIMITER ;
KD3-98849-sakshi@localhost> CALL reactangle();
Query OK, 1 row affected (0.102 sec)

KD3-98849-sakshi@localhost> SELECT*FROM temp;
+-----+-----+-----+
| value | square | cube_value |
+-----+-----+-----+
|      5 |      25 |          125 |
+-----+-----+-----+
1 row in set (0.005 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
KD3-98849-sakshi@localhost> DROP PROCEDURE IF EXISTS convert_temp;
Query OK, 0 rows affected, 1 warning (0.015 sec)
```

```
KD3-98849-sakshi@localhost> DELIMITER $$
```

```
KD3-98849-sakshi@localhost> CREATE PROCEDURE convert_temp()
```

```
    -> BEGIN
    -> DECLARE temp_degree DOUBLE DEFAULT 56;
    -> DECLARE Celsius DOUBLE;
    -> DECLARE Fahrenheit DOUBLE;
    -> SET
    -> Fahrenheit=temp_degree;
    -> SET
    -> Celsius=(Fahrenheit-32)*(5/9);
    -> SET
    -> Fahrenheit=(9/5)*(Celsius+32);
    -> SELECT TEMP_DEGREE,Celsius,Fahrenheit;
    -> END;
    -> $$
```

```
Query OK, 0 rows affected (0.089 sec)
```

```
KD3-98849-sakshi@localhost> DELIMITER ;
```

```
KD3-98849-sakshi@localhost> CALL convert_temp();
```

TEMP_DEGREE	Celsius	Fahrenheit
56	13.334399999999999	81.60192

```
1 row in set (0.042 sec)
```

```
Query OK, 0 rows affected (0.062 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
KD3-98849-sakshi@localhost> DELIMITER $$
KD3-98849-sakshi@localhost>
KD3-98849-sakshi@localhost> CREATE PROCEDURE sp_convert_inches()
-> BEGIN
->     DECLARE v_inches INT DEFAULT 124;
->     DECLARE v_yards INT;
->     DECLARE v_feet INT;
->     DECLARE v_rem_inches INT;
->
->     SET v_yards = v_inches / 36;
->     SET v_feet = (v_inches % 36) / 12;
->     SET v_rem_inches = v_inches % 12;
->
->     SELECT v_yards AS yards,
->           v_feet AS feet,
->           v_rem_inches AS inches;
-> END;
->
-> $$
Query OK, 0 rows affected (0.050 sec)

KD3-98849-sakshi@localhost>
KD3-98849-sakshi@localhost> DELIMITER ;
KD3-98849-sakshi@localhost> CALL sp_convert_inches();
+-----+-----+-----+
| yards | feet | inches |
+-----+-----+-----+
|      3 |     1 |       4 |
+-----+-----+-----+
1 row in set (0.020 sec)

Query OK, 0 rows affected (0.063 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```

KD3-98849-sakshi@localhost> DELIMITER $$
KD3-98849-sakshi@localhost>
KD3-98849-sakshi@localhost> CREATE PROCEDURE sp_product()
  -> BEGIN
  -> DECLARE num1 DOUBLE DEFAULT 10;
  -> DECLARE num2 DOUBLE DEFAULT 12;
  -> DECLARE product DOUBLE;
  -> SET product=num1*num2;
  -> IF product >=100 THEN
  -> SELECT num1,num2,product,'Product is equal to or greater than 100';
  -> ELSE
  -> SELECT num1,num2,product , 'Product is less than 100'AS result;
  -> END IF;
  -> END;
  -> $$
Query OK, 0 rows affected (0.121 sec)

KD3-98849-sakshi@localhost> DELIMITER ;
KD3-98849-sakshi@localhost> CALL sp_product();
+-----+-----+-----+-----+
| num1 | num2 | product | Product is equal to or greater than 100 |
+-----+-----+-----+-----+
| 10 | 12 | 120 | Product is equal to or greater than 100 |
+-----+-----+-----+-----+
1 row in set (0.020 sec)

Query OK, 0 rows affected (0.093 sec)

```